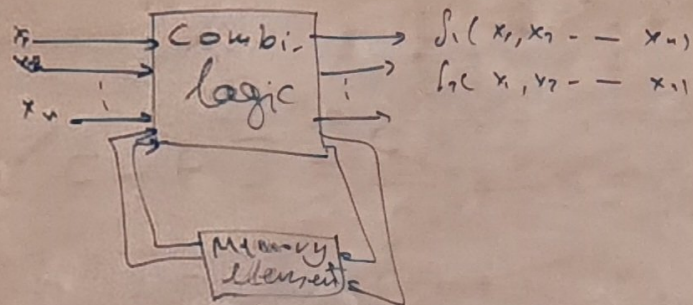
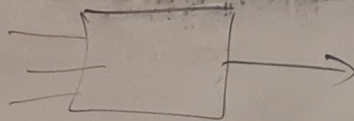


Combinational Logic

1. Design of Comb Logic



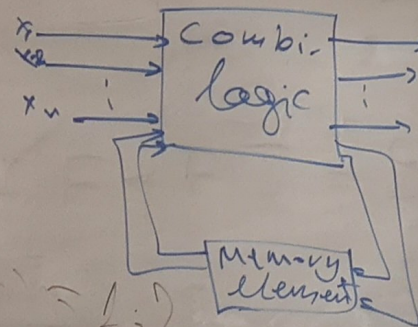
Combinational Logic



$$f = \bar{a}bc + a\bar{b}c + ab\bar{c} + abc$$

$\bar{a}$				
a				

a	b	c	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1



$$f_1(x_1, x_2, \dots, x_n)$$

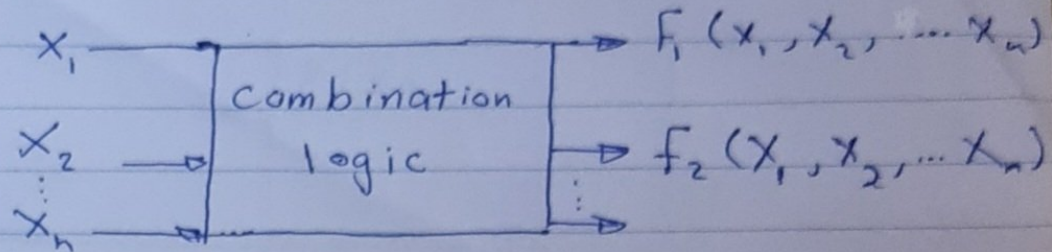
$$f_2(x_1, x_2, \dots, x_n)$$

1. Design of Comb. Logic

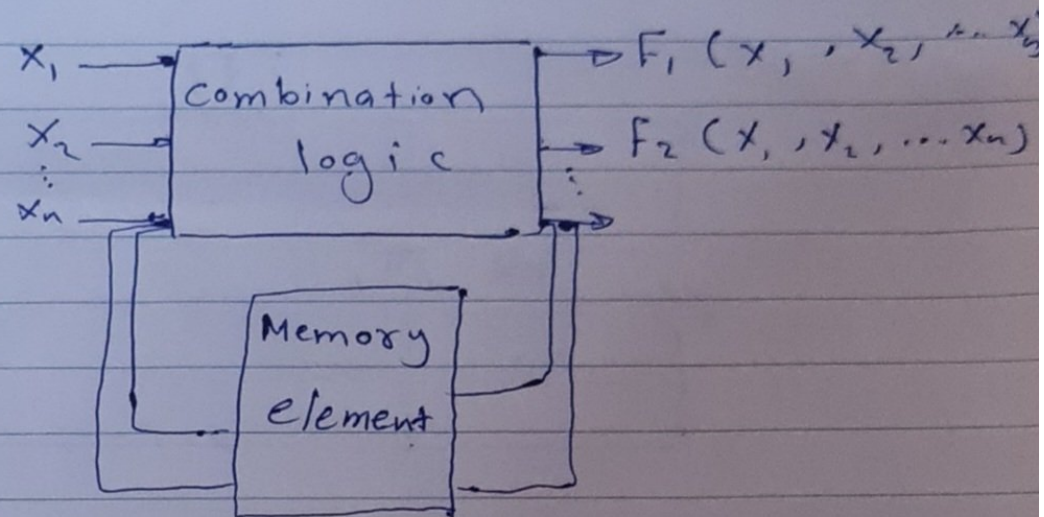
Ex: Design the Comb. Circuit with 3-input and 1-output: if the most input equal 1, the output $F = 1$

We have two main circuit in computer:

① Combinational



② Sequential



سؤال قد يأتي في الاختبار:

What is the different between combinational logic and sequential logic ? with drawing.

Design the comp circuit with 3 input and 1 output
if the most input equal 2, the output equal = 1, 2.

①

Solution	abc	F
	0 0 0	0
	0 0 1	0
	0 1 0	0
$\bar{a}bc$	0 1 1	①
	1 0 0	0
$a\bar{b}c$	1 0 1	①
$ab\bar{c}$	1 1 0	①
abc	1 1 1	①

② $\bar{a}bc + a\bar{b}c + ab\bar{c} + abc$

③ simplifying

	$\bar{b}\bar{c}$	$\bar{b}c$	bc	$b\bar{c}$
$\bar{a}$			①	
b		①	①	①

$F = bc + ac + ab$

④ drawing.

Q.

Design the combination circuit with 3 input and 1 output if the sum of inputs equal 2 and the output equal 2.